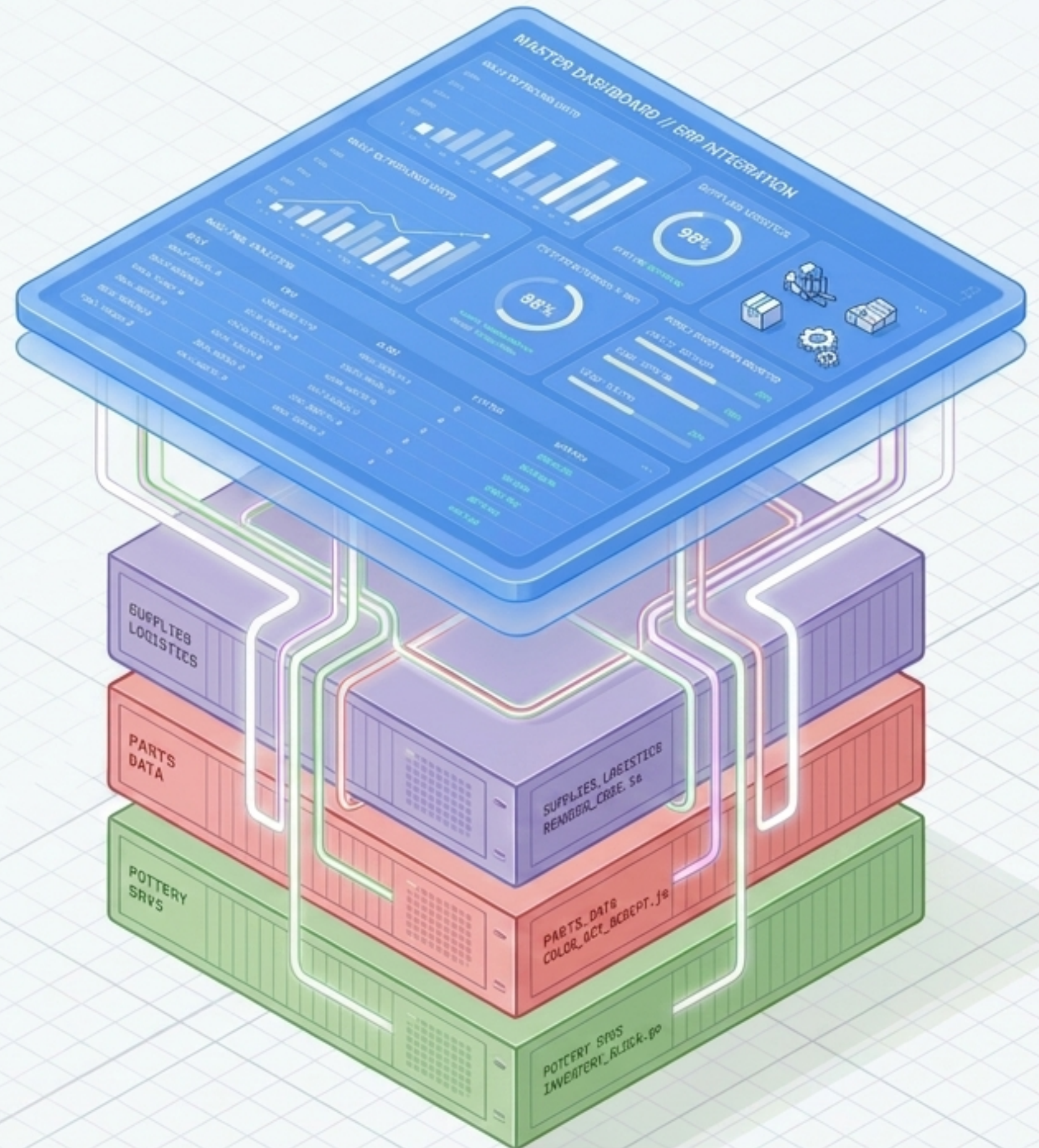


The Automated Warehouse Engine

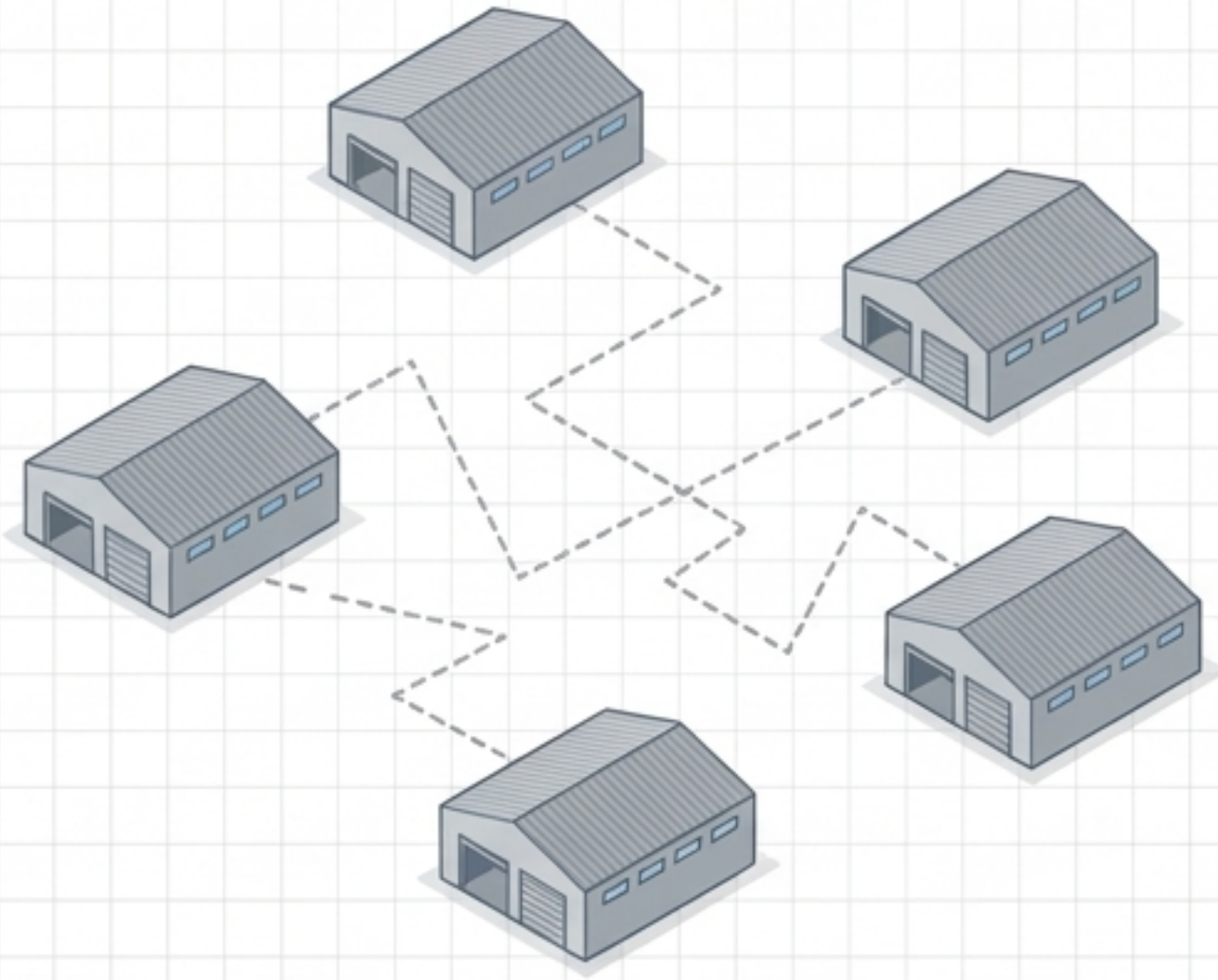
Translating custom Google Apps Script into a closed-loop ERP system.

GitHub Repository: [Evgenii_Byalo](#)



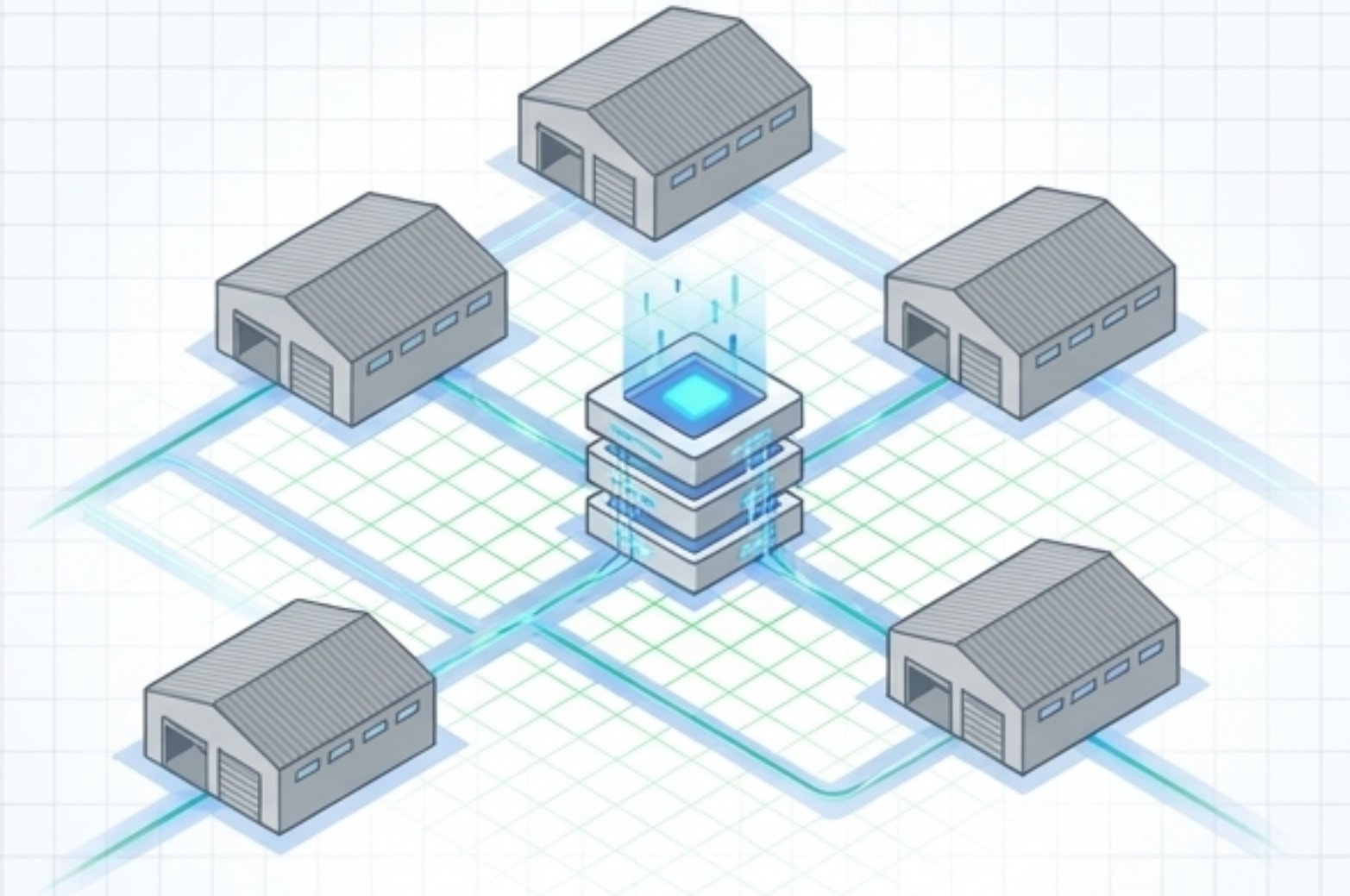
Fragmented Physical Operations Create Systemic Blind Spots

THE CHALLENGE



5 Remote Sites, Thousands of SKUs,
Manual Daily Tallies.

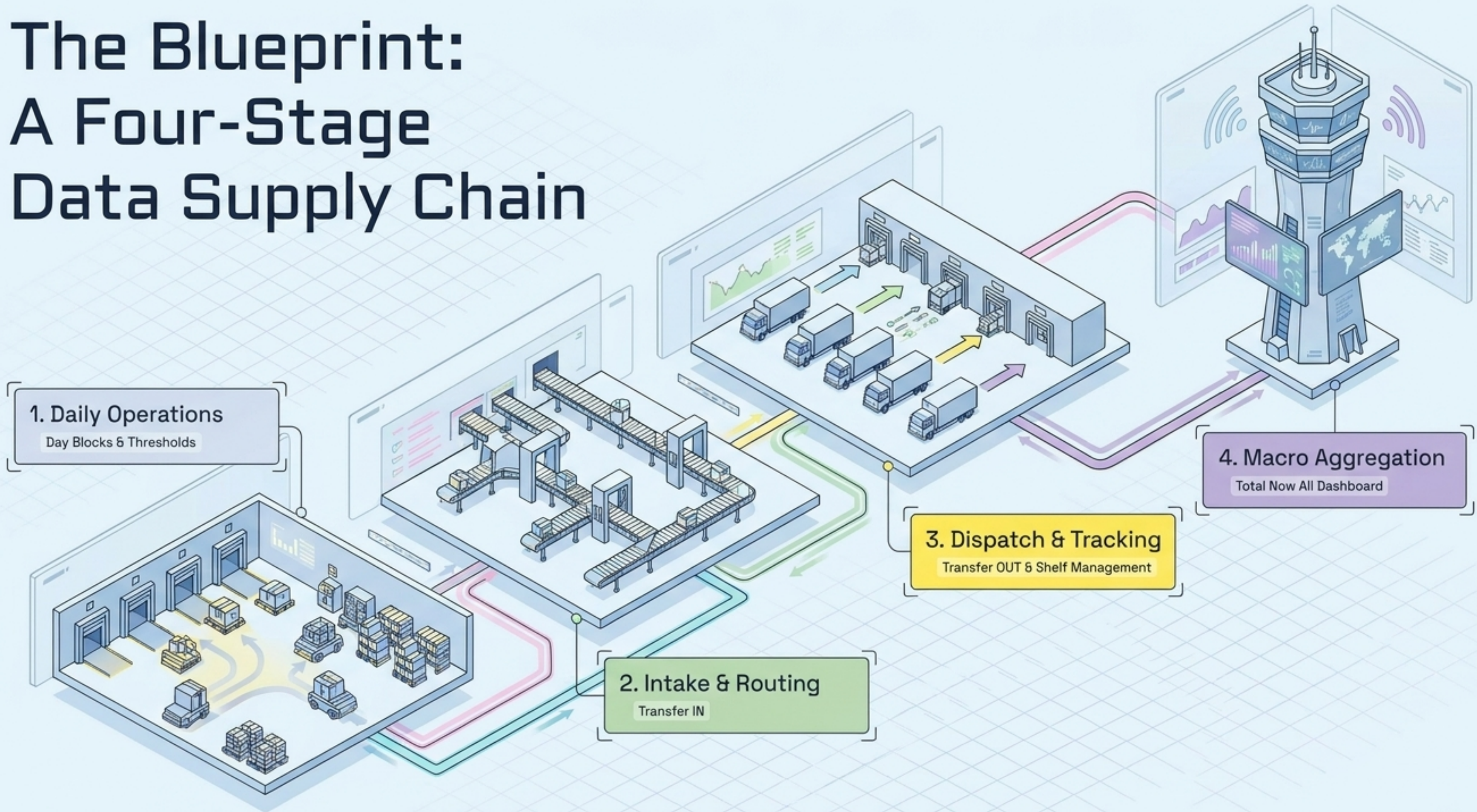
THE SOLUTION



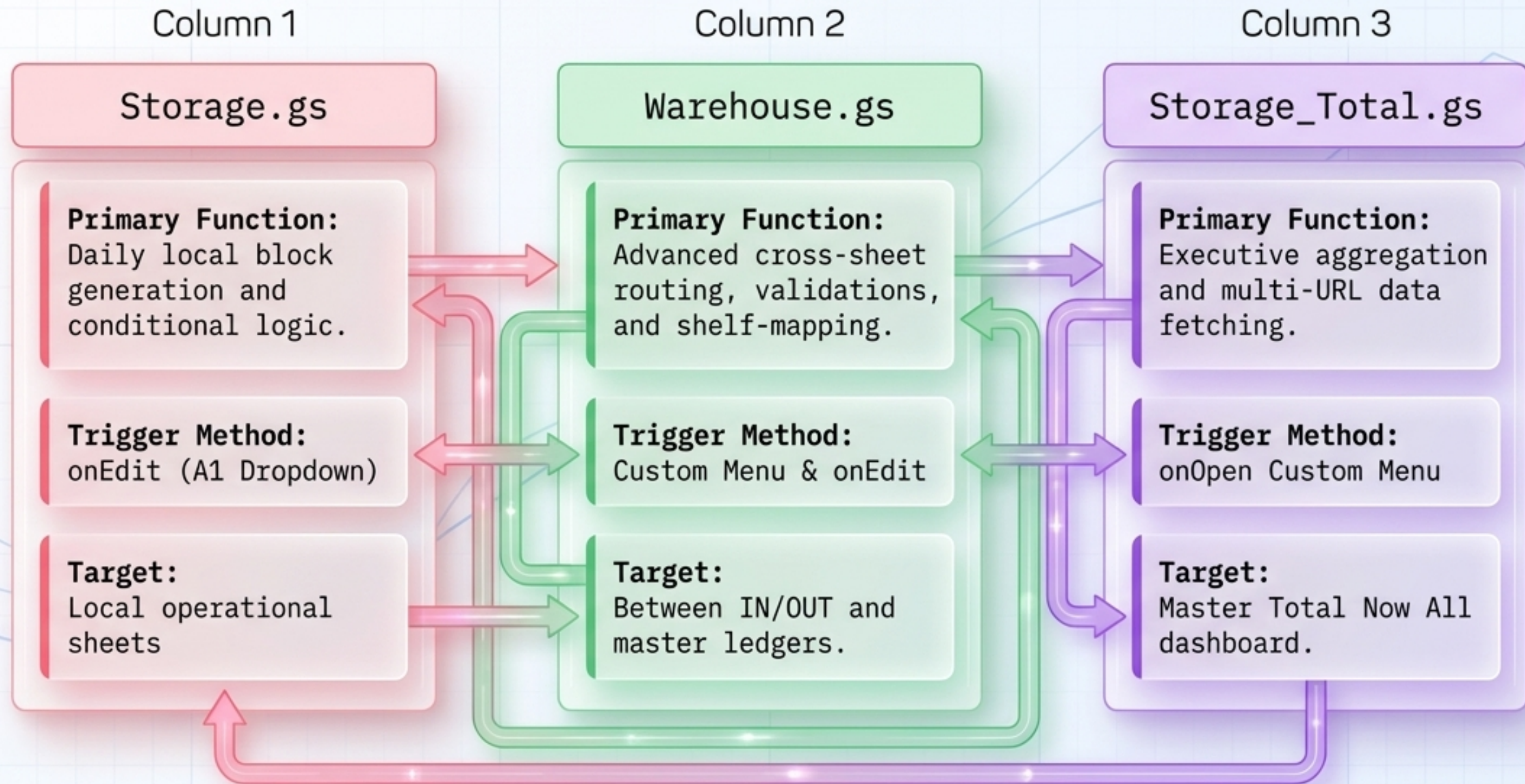
1 Unified Dashboard, Automated Daily
Blocks, Algorithmic Shelf Routing.

A custom spreadsheet ecosystem acting as a lightweight, real-time Enterprise Resource Planning (ERP) platform.

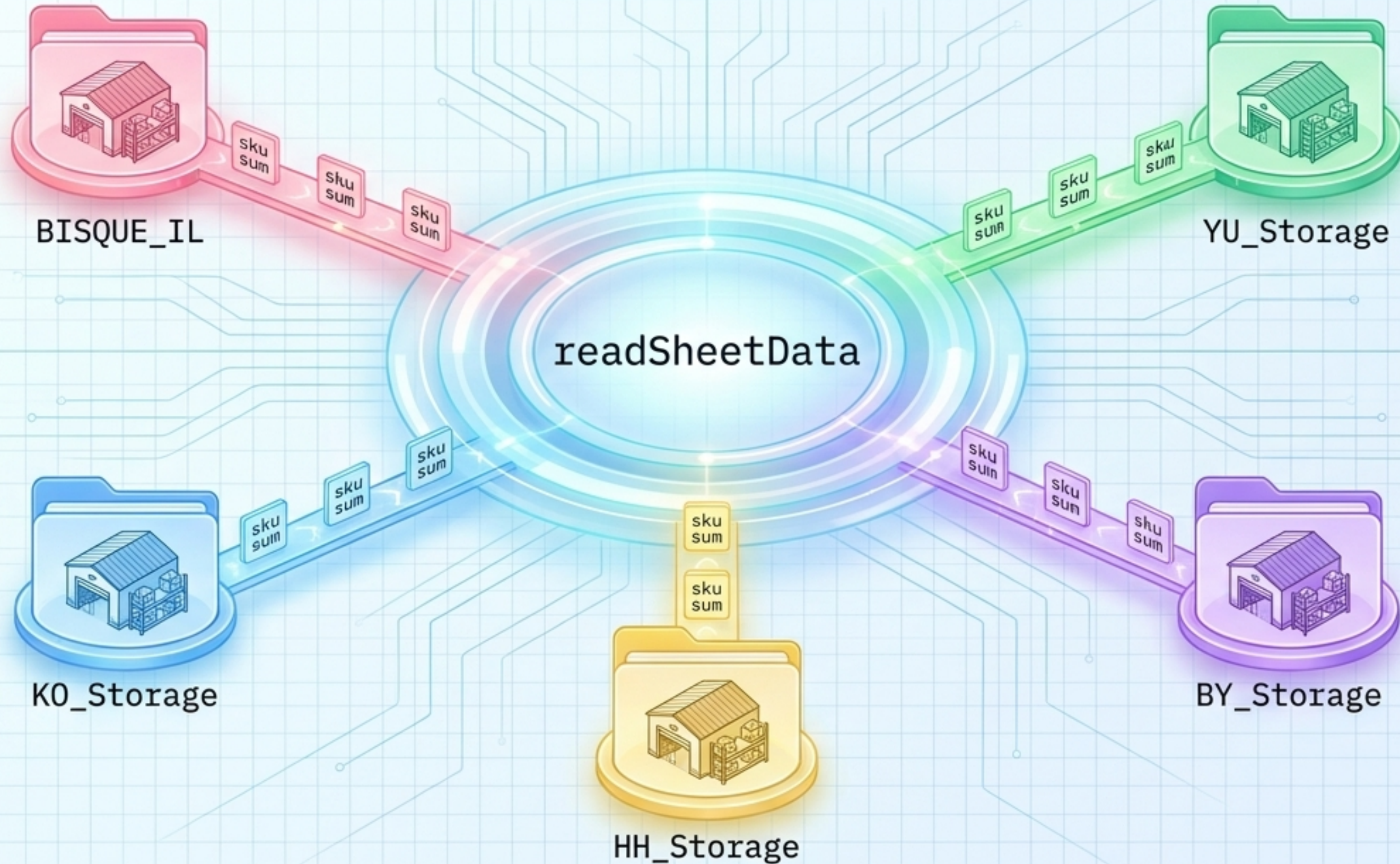
The Blueprint: A Four-Stage Data Supply Chain



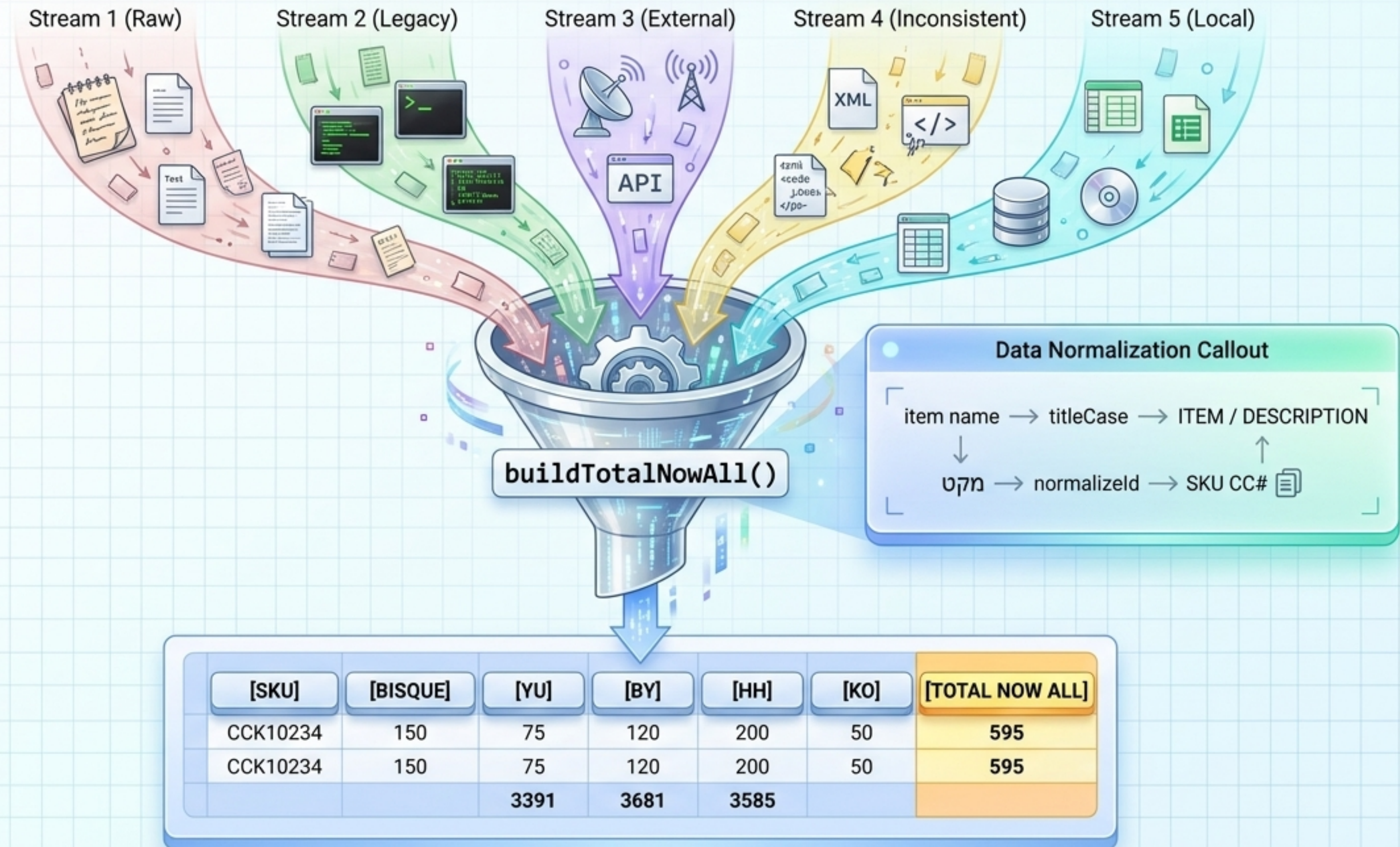
System Architecture Relies on a Three-Pillar Module Matrix



The Multi-Warehouse Network Fetches Remote Telemetry in Real-Time



Zero Manual Entry: Aggregating the Total Now All Master Array



Category Pipelines Enforce Distinct Tracking Rules by Material

POTTERY

- Complex tracking protocol. 
- Requires [Shelf #] and [BOXES PER MASTER CARTON] 
- Features automated A→Z sorting by Description column 

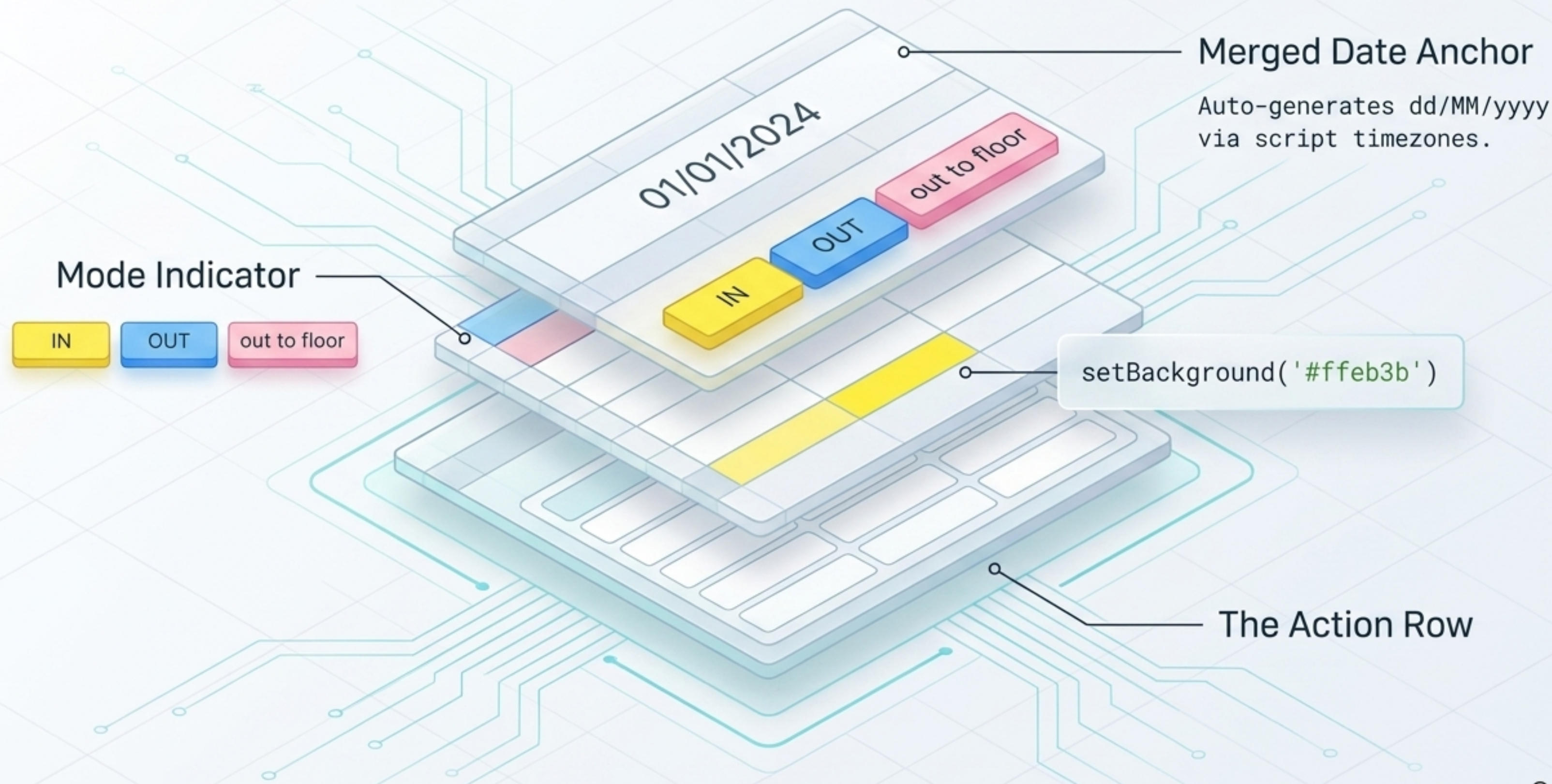
PAINTS

- Streamlined, high-velocity tracking. 
- Bypasses individual shelf and box tracking for raw volume management 

SUPPLIES

Standardized auxiliary pipeline supporting both primary streams.

The Anatomy of a Day Block: Bridging Time and Inventory

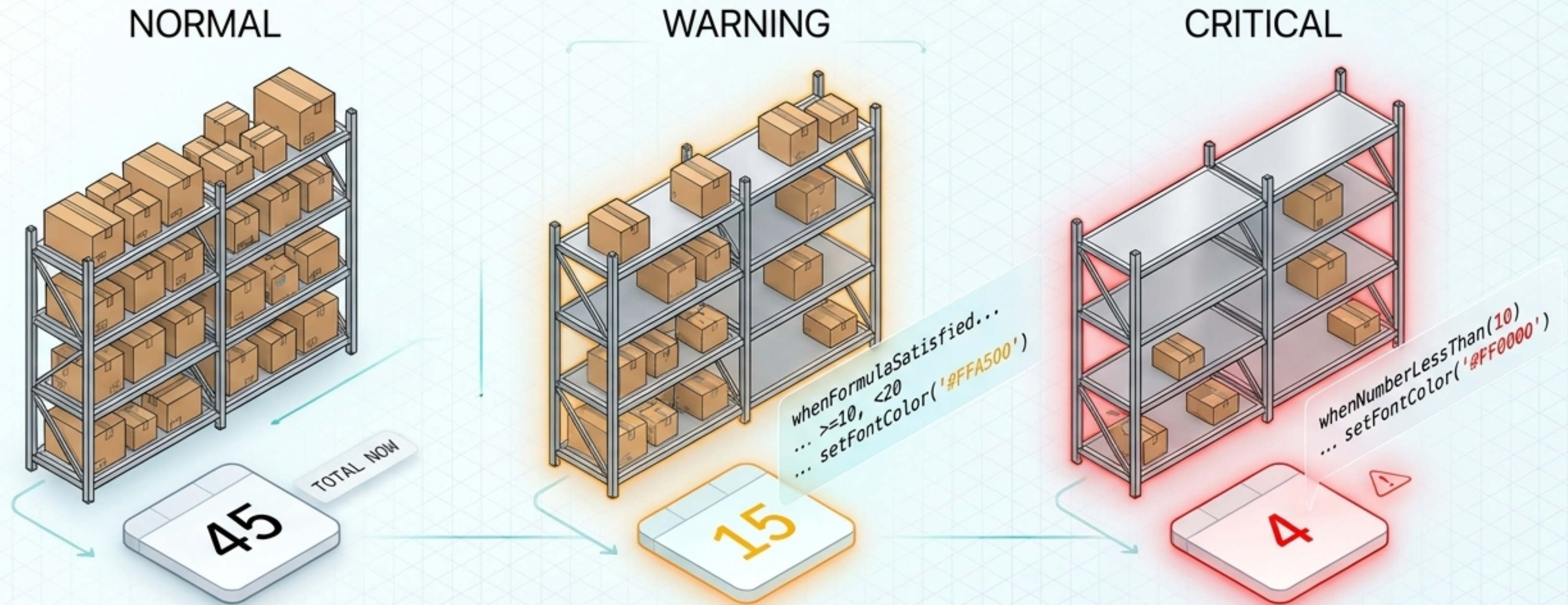


Algorithmic Injection Transforms Static Grids into Timelines



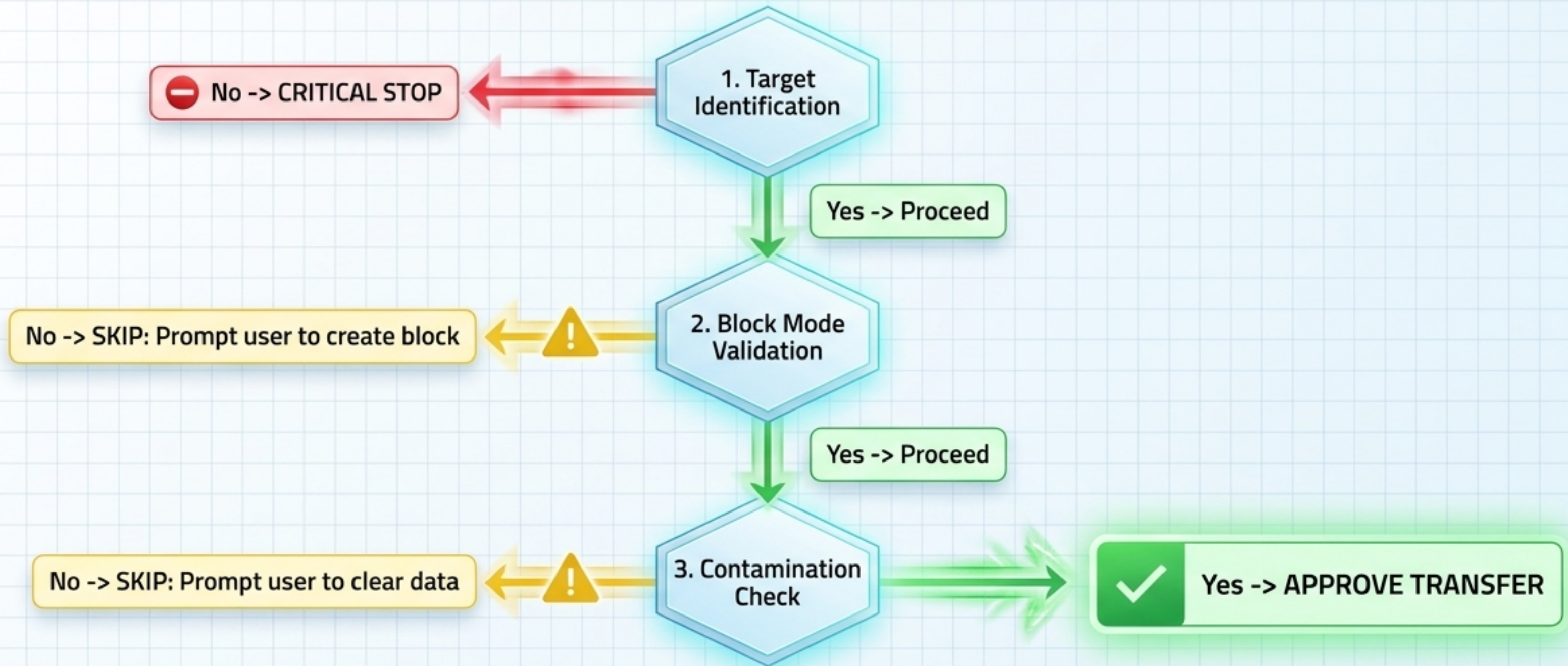
Executing **addNewDayBlock** creates a chronological timeline by instantly appending localized time units, seamlessly bridging yesterday's running total to today's physical movement.

The Automated Tripwire: Syncing Physical and Digital Vulnerabilities



`applyTotalNowConditionalFormatting` acts as an automated tripwire, instantly highlighting supply chain vulnerabilities as Day Blocks deduct items.

The Gatekeeper Protocol: Strict Pre-Transfer Validation



No data moves until the Gatekeeper validates all three safety checks.

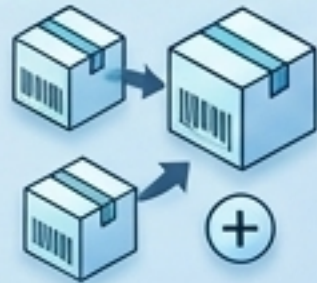
Workflow Matrix: The Rules of Intake vs. Dispatch

Transfer IN (Receiving)

Target Sheet Check:
Validates for empty IN block.



SKU Handling:
Aggregates identical SKUs;
generates new rows for
unknown SKUs.



Metadata:
Captures [Boxes per Master
Carton] and [Shelf #].

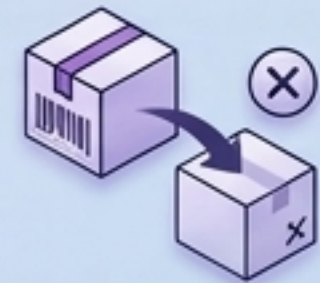


Transfer OUT (Dispatch)

Target Sheet Check:
Validates for empty OUT block.



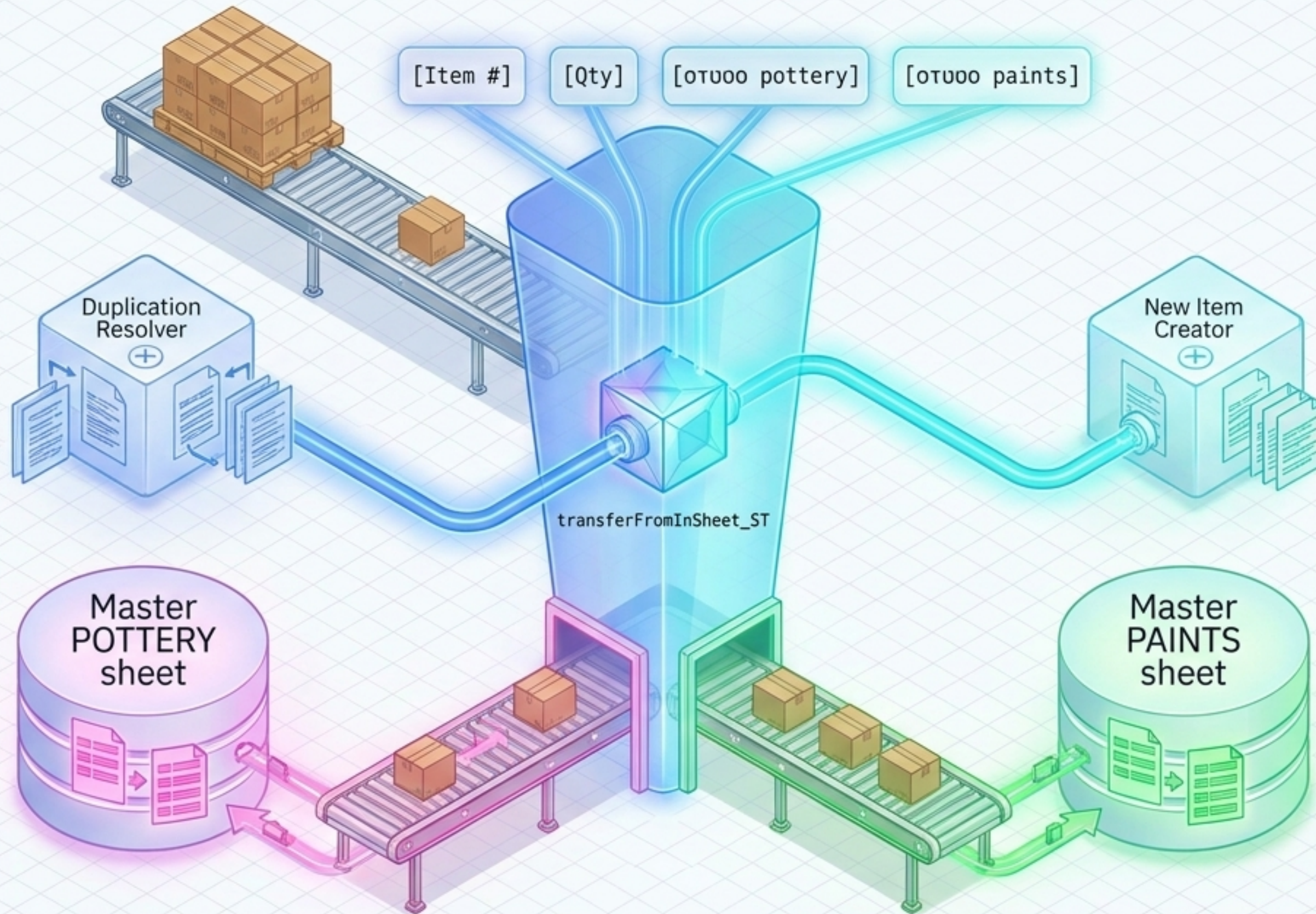
SKU Handling:
Deducts from existing
inventory; ignores
unrecognized SKUs.



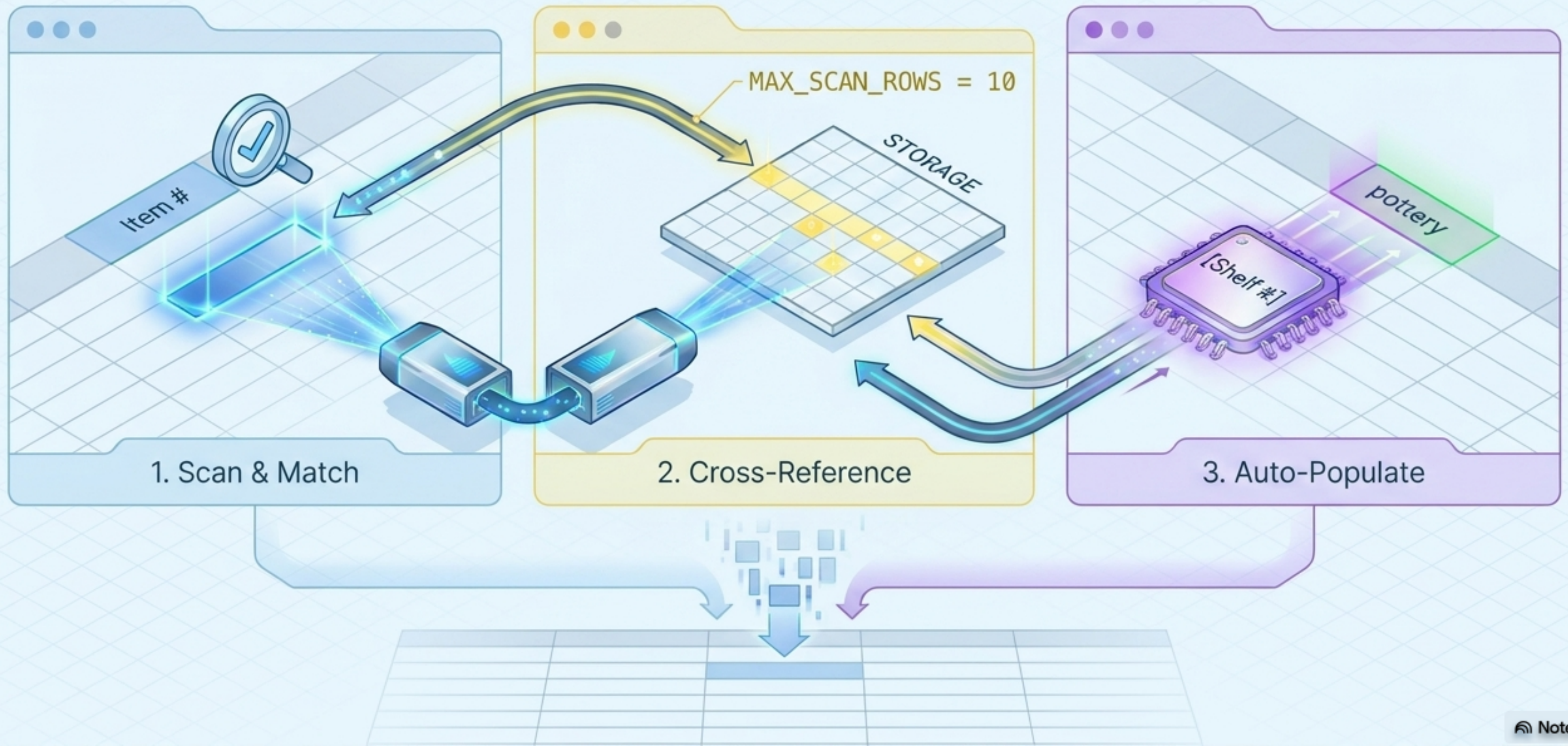
Metadata:
Pulls [Shelf #] from master
storage for physical picking.



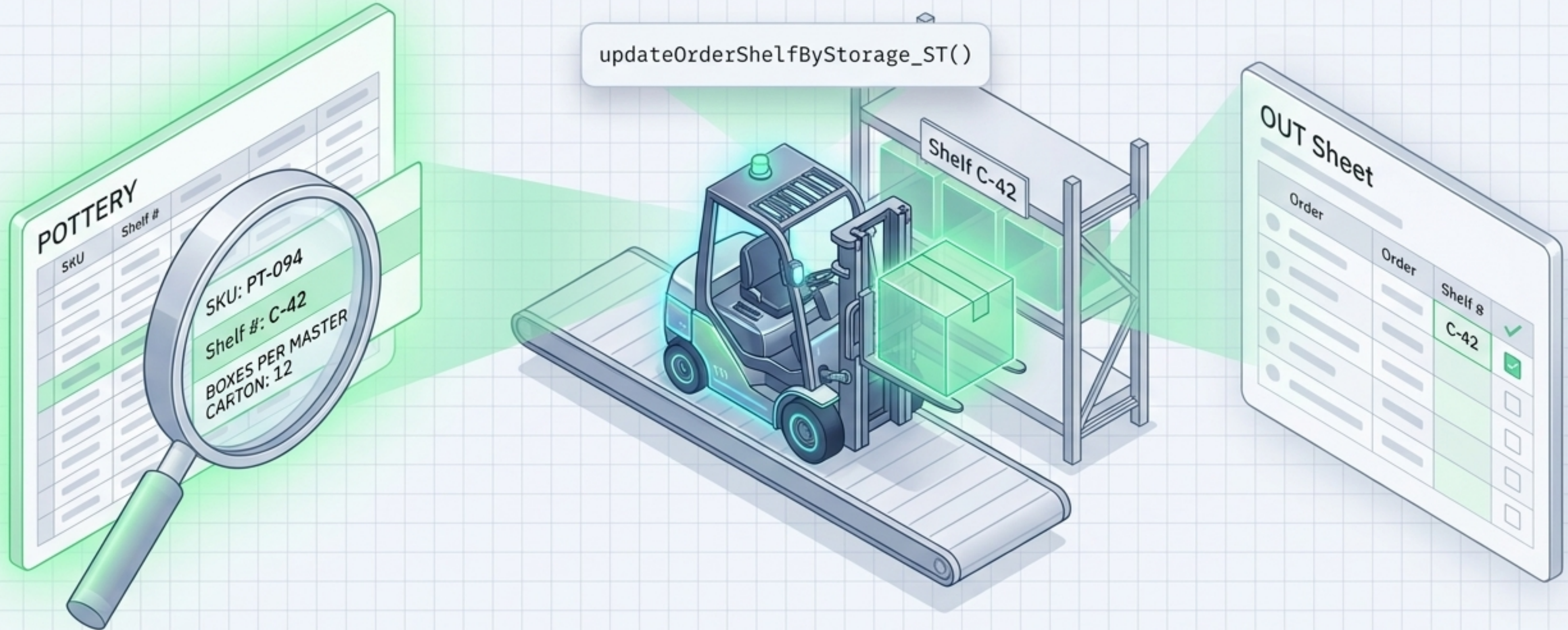
Routing Intelligence: The Intake Pipeline



Routing Automation: The Shelf-Mapping Algorithm



Dispatch Execution: The Digital Forklift



By cross-referencing master storage arrays, the system auto-populates exact warehouse coordinates directly onto the fulfillment team's checklist, eliminating physical search time.

Synthesis: A Closed-Loop Inventory Ecosystem



Beyond Spreadsheets: Forging a Living ERP

```
function validateData(row) {  
  return row.normalizeId().titleCase();  
}  
  
const syncPhysical = async (data) => {  
  await database.update(data.SKU);  
}  
  
//  
macroAggregation =  
  stream1.concat(stream2, stream3, stream4)  
  .filter(isValid);  
  
// the data need to sync preventive item names  
const syncPhysical = async (data) => {  
  await database.update(data.SKU);  
}  
  
// functions from mascara...
```



Intelligent
Validation



Physical-Digital
Sync



Automated
Oversight



By combining dynamic array aggregations, chronological block-building, and localized shelf intelligence, raw script actively manages the warehouse, allowing operators to focus on movement, not mathematics.